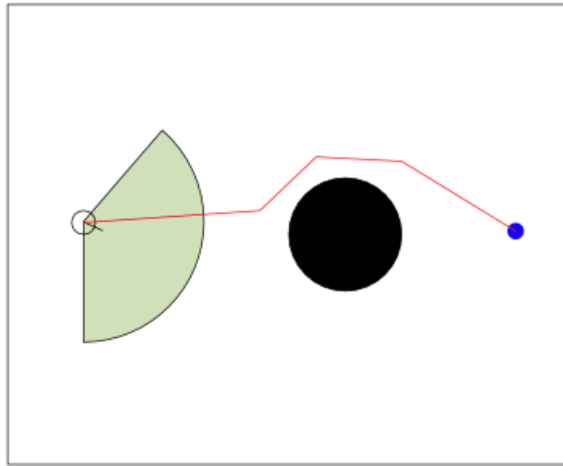


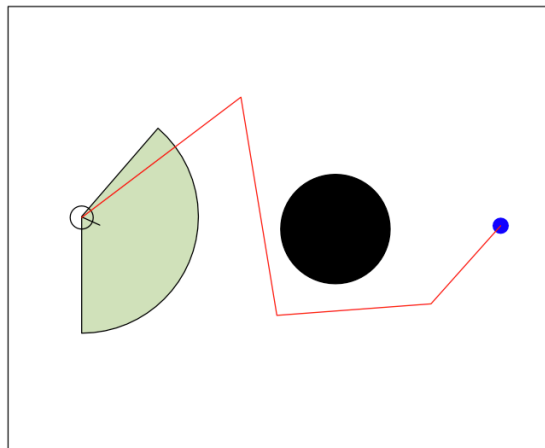
## Research statement

My research focuses on agent-based simulation and modeling of steering behaviors. Basically, the studied models need to have a set of behaviors that make agents move around and fulfilled requirements. For example, in the figure below, the agent is the hollow circle, the line from the center is the current heading direction, the green arc is its filed of view, the solid black circle is an obstacle, and the blue small circle is the position that the agent needs to reach.

As you can see, the red path presents a solution of a model for the agent to reach the blue dot.

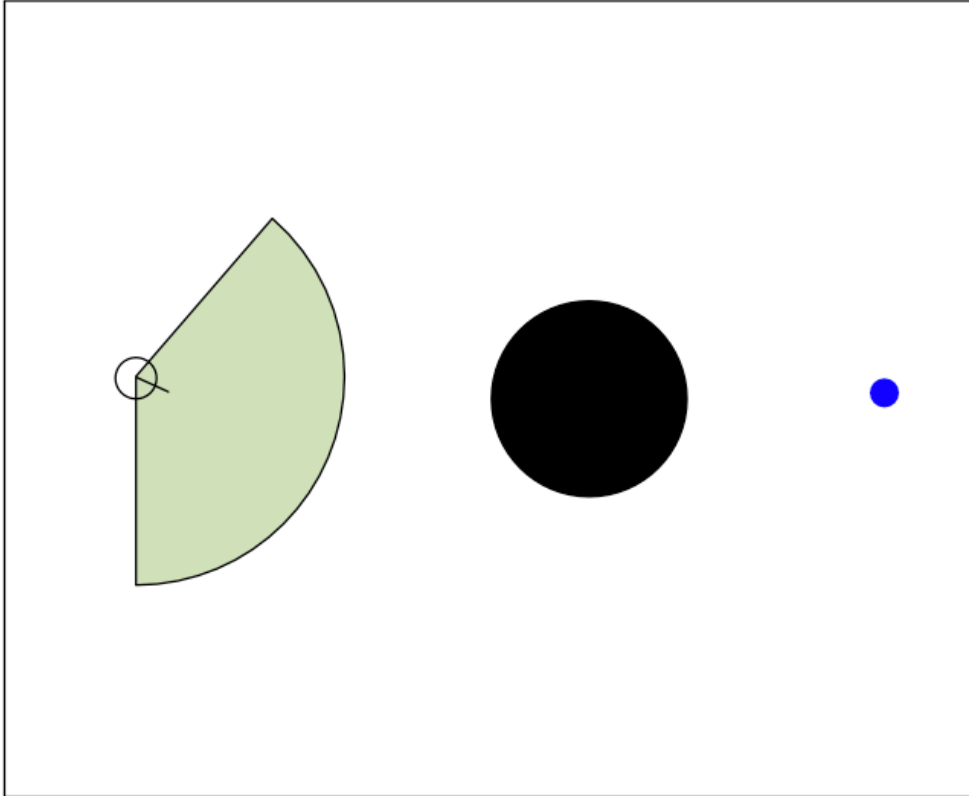


Of course, the solution above is just one of them. You can have countless other models as well. For example, the figure below is another approach.



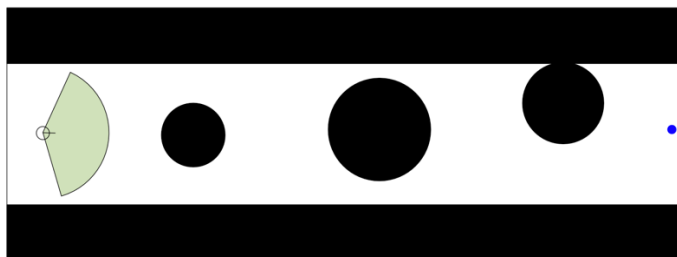
To discover the best model, one modeler like me needs to carefully design the model, but it contains human bias. In this research, I would like one student to use AI and let the algorithms discover the model and compared with what I already have. The objective is very straight forward.

1. Use a image as the input. (Image below is one example)
2. Use any AI approaches to find the red path.



Please craft a proposal and show what you are planning to do the achieve these objectives. The proposal should:

1. Describe the approach or approaches. Think of a general solution that it should work with other layouts such as the one below.



2. What challenges you think you might get.
3. What questions you have for me.

**Due:** Before August 1<sup>st</sup>, 2026, however, since there is more than one student, whoever sends it sooner might have an advantage. For example, if the first student provides a well-done proposal, there is no need for me to wait for the rest.